

26TH DIGITAL BLUE ECONOMY SUMMIT

# ParaSail

A smart guide for safer fishing and healthier oceans

*Predictive Advisory and Retrieval-Augmented System Advancing Informed Livelihoods*

Is it safe to sail?

Are the fish there?

Is it allowed?

One simple answer to all three questions - before you leave the harbour.

Anantha Krishnan AS · Naipunnaya School of Management, Cherthala, India · September 2026

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One clear answer: GO, GO CAREFULLY, WAIT OR MOVE, STOP

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The answer in plain words - with its reasons listed

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Fishers, the ocean, communities, authorities

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One simple idea, and what comes next

*Tap any card to jump straight to that part of the story.*

# Why fishing needs help

1 in 3

**fish populations are now overfished**

FAO, 2023

3× more

**plastic flowing into the ocean by 2040**

UNEP, 2023

500 million

**people depend on small-scale fishing**

FAO, 2023

## ◆ Fish are moving

Warming seas push fish to new waters, and the old seasonal calendars drift out of step.

## ◆ Spawning is disturbed

Fishing while fish breed leaves fewer fish for everyone next year.

## ◆ The sea is dangerous

District-wide forecasts say nothing about the wind and waves off your own ground.

## ◆ Rules are hard to check

Boundaries and closed seasons live in offices and on paper - not at sea.

# Meet ParaSail - one clear answer

## Like a weather app - but for fishing.

You tell it where you will fish, what you are catching and when. Before you leave the harbour it gives you one simple answer - and the reasons behind it.

Free · in your language · installs like a phone app

*Click to reveal each signal*



### GO

Calm seas, fish likely, zone open - safe to sail.



### GO CAREFULLY

Fishing is possible - but watch the changing weather.



### WAIT OR MOVE

Rough seas or young fish nearby - wait, or try another spot.



### STOP

Not safe, or not allowed. Stay ashore today.

**Protected area or closed season?** The answer is always STOP - whatever the weather.

# What you actually see

TOMORROW, 5 AM — OFF KOCHI

# GO

## PLAIN-LANGUAGE EXPLANATION

- ◆ GO — calm seas.
- ◆ Sea right now: wind 12 km/h, waves 0.8 m, currents 1.2 km/h.
- ◆ Indian oil sardine likely in this area.
- ◆ Risk to young fish and habitats: low.

*Every point is a separate reason you can check - not a wall of text.*

WHILE IT THINKS



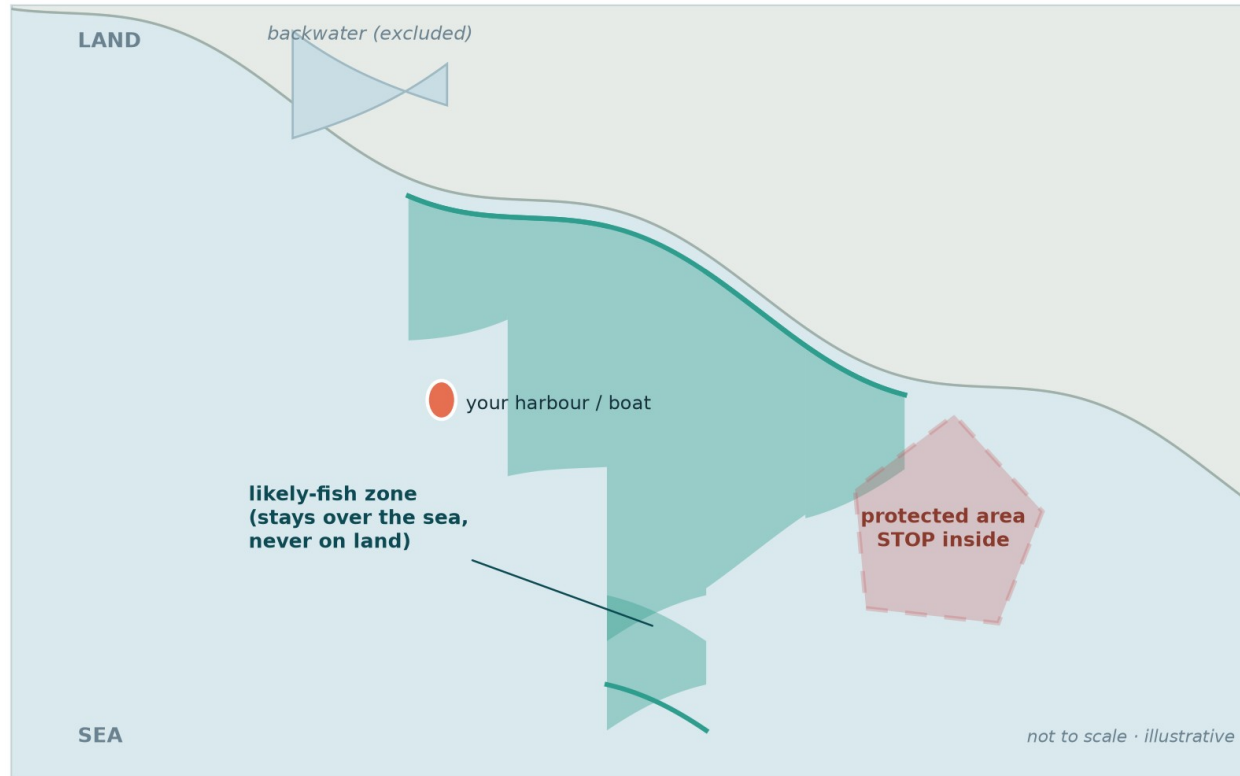
## Reading the sea...

A live progress cue - animated wave bars - so you always know it is working.

**Nothing is invented.** If the data cannot support a claim, the system says so instead of guessing - and the full detail stays one tap away.

**Simple on purpose.** A fishers' eye view first; the numbers live behind it for anyone who wants them.

# The map shows where - honestly



*Illustrative view of the coast, the likely-fish zone and a protected area.*

**1**

## Follows the real coastline

The likely-fish area is drawn from actual shoreline geometry - it never spills over land or into backwaters.

**2**

## Protected areas, for real

Real marine-park boundaries. Inside one the answer is STOP - and it names the park you are in.

**3**

## Honest when nothing fits

If conditions do not suit the fish, it shows a small honest zone - or none - instead of inventing one.

*A fishing zone that covers a town, or a 'protected area' you cannot see, protects nothing. The map is built so a fisher can trust it.*

# Ask it anything - in your own words

## ASK PARASAIL

"Can I take my small boat out tomorrow?"

GO CAREFULLY - waves near 1 m in the morning, wind rising after noon. Best back before two.

*sources: marine forecast · rulebook*

"And in June, for sardine?"

1

### Answers in plain words

Ask like you would ask a neighbour - and it replies in your own language.

2

### Shows its sources

Every answer points to the forecast, rule or local knowledge it used.

3

### Honest about the hardware

It never pretends a model is running when it is not.



**AI: qwen2.5vl-3b-laptop · READY**

It checks what is actually installed on this computer and says so - and if a model is missing, it prints the exact command to get it.

*The AI explains the advice - it never overrules it. Closed season? Still STOP.*

# It worked - here is what we found

- ✓ **Live data kept flowing** Even when the internet dropped, advice continued - clearly marked as slightly older information.
- ✓ **Closed seasons were always respected** Every single test during a closure came back STOP. No exceptions, ever.
- ✓ **Advice changed sensibly** When the weather turned or the fish moved, the traffic light changed exactly as it should.
- ✓ **Every answer explained itself** The reasons shown next to each advice were checked - and they were the right ones.
- ✓ **The map stayed over water** Likely-fish zones followed the coast and never covered land, lakes or backwaters.
- ✓ **The AI never overruled the verdict** It explains, cites and defers - the decision stays with the rules, not the model.

# Everyone on the coast wins

## ◆ Fishers

Save fuel, stay safer, and avoid fines from honest mistakes.

## ◆ The ocean

Protected zones and spawning seasons are left in peace - automatically.

## ◆ Communities

Generations of fishing knowledge kept in a lasting digital library.

## ◆ Authorities

Clear, fair decisions backed by evidence anyone can see.

## Made for your coast

Quick to set up for a new coast · runs on an ordinary computer · installs like a phone app on the boat · answers in your language, local fish names included

*Next: a season-long trial with fishing communities, and voice guidance for hands-free use at sea.*

# Fish smarter. Stay safe. Keep the ocean giving.

ParaSail answers three questions every fisher already asks - is it safe, are the fish there, is it allowed - and shows its reasons every single time.

*The phone app is here today - installed from the website, it works offline and speaks your language. Next: a season-long trial with fishing communities, and voice guidance.*

Aligned with the UN Sustainable Development Goals: 14 Life Below Water · 2 Zero Hunger · 8 Decent Work · 9 Innovation · 13 Climate Action

## Thank you

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